

NIFTY 100 ANALYST GUIDE

Financial Analytics, Screening, Valuation, Risk and Company Intelligence Platform

Purpose: This guide explains how an analyst can use the Nifty 100 analytics platform to review companies, compare sectors, evaluate valuation, assess risk, inspect cash-flow quality and access generated company reports.

1. Platform Overview

The platform combines structured financial data, financial ratio analysis, stock screening, peer analytics, valuation analysis, risk analysis, NLP-based insights, company tearsheets, sector reports, portfolio statistics, an interactive dashboard and a REST API.

2. Data Foundation

- SQLite database: **data/nifty100.db**
- Company master information
- Profit and loss statements
- Balance sheets
- Cash-flow statements
- Financial ratios
- Market capitalization and valuation metrics
- Sector and peer-group information
- Historical stock prices
- Pros and cons and NLP-derived company insights

3. Company Analysis

Use the company scorecard as the primary company-level starting point. It combines profitability, capital efficiency, cash-flow quality, leverage and market information.

Metric	Purpose
ROE	Measures return generated on shareholders' equity.
ROCE	Measures efficiency of capital employed.
Net Margin	Shows profitability after expenses.
Operating Margin	Shows operating profitability.
Debt to Equity	Indicates financial leverage.
Interest Coverage	Indicates ability to service interest costs.
P/E	Market valuation relative to earnings.
P/B	Market valuation relative to book value.
Dividend Yield	Cash dividend yield relative to market price.

4. Stock Screener

The stock screener allows analysts to narrow the Nifty 100 universe using financial and valuation criteria.

- Profitability and return metrics
- Leverage and financial strength
- Valuation metrics
- Market capitalization
- Quality-oriented ranking
- Sector-based filtering

5. Valuation Analysis

Valuation analysis combines P/E, P/B and ROE ranking to identify companies with relatively attractive valuation characteristics. The system produces top valuation picks, undervalued companies and overvalued companies.

Important: The valuation outputs are analytical screening results and should not be interpreted as investment recommendations by themselves.

6. Risk Analysis

- Debt-to-equity
- Interest coverage
- Financial leverage
- Lowest-risk company ranking
- Highest-risk company ranking
- Risk summary

7. Cash-Flow Intelligence

Cash-flow intelligence evaluates operating, investing and financing cash-flow information and produces a cash-flow health classification, risk level, commentary and score.

Output: **output/cashflow_intelligence.xlsx** and **output/cashflow_insights.csv**.

8. NLP and Company Intelligence

The platform contains NLP-oriented outputs including company pros and cons, parsed historical analysis metrics and generated company intelligence.

Relevant outputs include **pros_cons_generated.csv** and **analysis_parsed.csv**.

9. Peer Radar Analysis

Radar charts compare an individual company's ROE, ROCE, net margin and quality characteristics against its broad sector average.

Generated charts are stored under **reports/radar_charts/**.

10. Company Tearsheets

A PDF tearsheet is generated for each company. The tearsheet provides a compact company-level analytical view and can be used as a quick research reference.

Location: **reports/tearsheets/**

The REST API also exposes company tearsheet retrieval.

11. Sector Analysis

Sector analysis aggregates company-level scorecard information to provide sector-level profitability, leverage, valuation, market-capitalization and quality comparisons.

Generated sector reports are stored under **reports/sector/**.

12. Portfolio Analysis

Portfolio analytics provide aggregate portfolio-level statistics and capital-allocation analysis. The generated portfolio report is available under **reports/portfolio/**.

13. Interactive Dashboard

- Home dashboard
- Company profiles
- Stock screener
- Sector analysis
- Valuation analysis
- Risk analysis
- Capital allocation
- Portfolio statistics
- NLP insights
- Company reports

Launch command:

```
streamlit run src/dashboard/app.py
```

14. REST API

The project provides a FastAPI-based REST API for accessing company, screening, valuation, peer, report and portfolio analytics.

Application entry point: **src/api/main.py**.

API documentation is available through the generated OpenAPI specification in **docs/openapi.json**.

15. Quality Assurance

Run the complete automated test suite from the project root:

```
python -m pytest -q
```

A successful run should report all collected tests as passing.

16. Recommended Analyst Workflow

1. Start with the company profile and basic company information.
2. Review the company scorecard and profitability metrics.
3. Compare the company with its sector peers.
4. Review valuation using P/E, P/B and ROE.
5. Review leverage and interest coverage.
6. Inspect cash-flow intelligence.
7. Review the company tearsheet.
8. Use sector reports for broader industry context.
9. Use the screener to identify comparable companies.
10. Use the dashboard or REST API for interactive/external access.

17. Interpretation Notes

- Higher ROE and ROCE generally indicate stronger capital efficiency.
- Higher margins generally indicate stronger profitability.
- Higher debt-to-equity generally indicates greater leverage.
- Higher interest coverage generally indicates stronger debt-servicing capacity.
- Lower valuation multiples can indicate relatively lower valuation, but must be considered alongside business quality and growth.
- Sector averages should be interpreted within the context of the sector's business model.
- Missing data should be treated as unavailable rather than automatically interpreted as zero.
- Analytics outputs are decision-support tools and are not standalone investment recommendations.

18. Key Output Locations

Output	Location
Company scorecard	output/company_scorecard.csv
Sector summary	output/sector_summary.csv
Valuation outputs	output/top_companies.csv
Risk outputs	output/risk_summary.csv
Cash-flow intelligence	output/cashflow_intelligence.xlsx
Company tearsheets	reports/tearsheets/
Sector reports	reports/sector/
Portfolio report	reports/portfolio/
Radar charts	reports/radar_charts/
Analytics report	output/analytics_report.txt
API specification	docs/openapi.json

End of Analyst Guide